Lab Report: Dynamic Routing Configuration using RIPv2

Activity by **IT Dose**

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This report documents the deployment, optimization, and verification of dynamic routing using RIPv2 across a multi-router Cisco Packet Tracer topology. The lab involved verifying initial local gateway connectivity, diagnosing inter-subnet reachability failures, configuring classless routing with passive-interface controls across routers R1 through R4, and confirming full end-to-end ICMP convergence.

1 Topology Overview & Addressing Scheme

This laboratory assignment focuses on implementing dynamic routing across an enterprise network using the **Routing Information Protocol Version 2 (RIPv2)**. The underlying network infrastructure, IP scheme allocation, and host end-device default gateway parameters follow the baseline architecture established by **IT-Dose**.

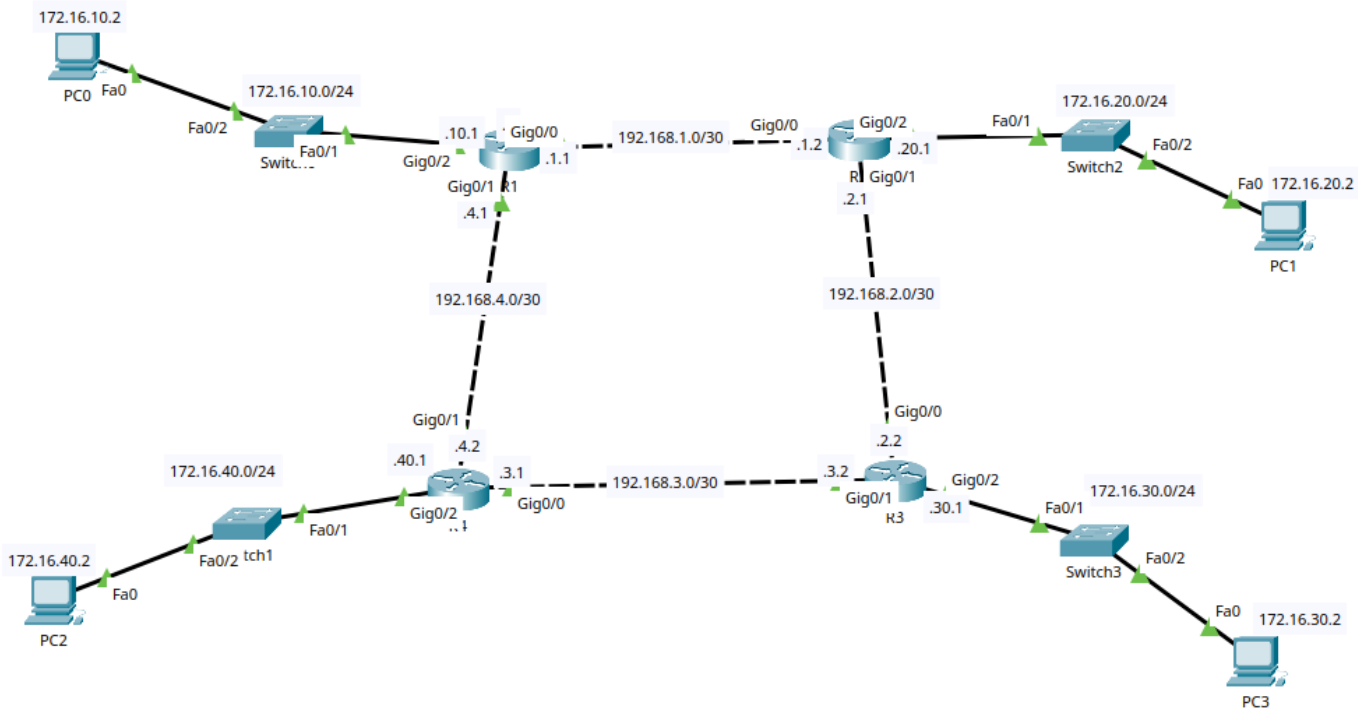


Figure 1: Cisco Packet Tracer Network Topology Diagram.

1.1 Subnet Allocation Summary

The topology consists of four distinct Local Area Network (LAN) segments interconnected via four point-to-point serial/Gigabit Ethernet WAN links:

Segment / Host	Subnet Prefix	Gateway Interface	Assigned Host IP
LAN 0 (PC0)	172.16.10.0/24	R1 (`Gig0/2`): 172.16.10.1	172.16.10.2
LAN 1 (PC1)	172.16.20.0/24	R2 (`Gig0/2`): 172.16.20.1	172.16.20.2
LAN 2 (PC3)	172.16.30.0/24	R3 (`Gig0/2`): 172.16.30.1	172.16.30.2
LAN 3 (PC2)	172.16.40.0/24	R4 (`Gig0/2`): 172.16.40.1	172.16.40.2
WAN Links	192.168.1.0/30 – 4.0/30	Inter-Router Connections	N/A

Table 1: Network Addressing and Subnet Mapping

2 Initial Network Connectivity Diagnostics

2.1 Phase 1: Local Default Gateway Verification

Before configuring dynamic routing protocols, host reachability to local default gateways was tested via ICMP echo requests. Because these links are directly connected, all hosts successfully communicated with their default gateways.

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC2	R4	ICMP	
	Successful	PC0	R1	ICMP	
	Successful	PC1	R2	ICMP	
	Successful	PC3	R3	ICMP	

Figure 2: Successful ICMP ping results between host PCs and local Router Gateways.

2.2 Phase 2: Pre-Configuration Inter-Subnet Reachability Failure

Attempts to communicate between PCs across different subnets resulted in failures. Routers lacked routing table entries for non-directly connected networks, dropping cross-subnet traffic.

Fire	Last Status	Source	Destination	Type	Color
	Failed	PC0	PC3	ICMP	
	Failed	PC2	PC1	ICMP	
	Failed	PC3	PC1	ICMP	
	Failed	PC2	PC0	ICMP	
	Failed	PC1	PC0	ICMP	

Figure 3: Failed ICMP pings across remote host destinations prior to dynamic routing deployment.

3 RIPv2 Protocol Configuration

To establish global reachability, **RIPv2** was enabled on all core routers. The configuration enforces classless routing update propagation, disables automatic classful network summarization, advertises local networks, and secures LAN interfaces using passive interface rules.

3.1 Cisco IOS Command Sequence

The standard command set applied across the routers (demonstrated below on generic router template `RX`) is structured as follows:

```
Router> enable
Router# configure terminal
Router(config)# hostname RX

! Step 1: Initialize RIP process and force Version 2
RX(config)# router rip
RX(config-router)# version 2
RX(config-router)# no auto-summary
```

```
! Step 2: Advertise participating local networks
RX(config-router)# network 172.16.X.0
RX(config-router)# network 192.168.X.0
RX(config-router)# network 192.168.X.0

! Step 3: Optimize bandwidth & security via passive-interface rules
RX(config-router)# passive-interface default
RX(config-router)# no passive-interface Gig0/0
RX(config-router)# no passive-interface Gig0/1
RX(config-router)# end
RX# copy running-config startup-config
```

Listing 1: Cisco IOS RIPv2 Configuration Commands

Design Note on Passive Interfaces: Deploying `passive-interface default` prevents RIP multicast traffic from saturating end-user host links.
Executing `no passive-interface` on inter-router links (`Gig0/0`, `Gig0/1`) allows control-plane neighbor adjacencies strictly over point-to-point connections.

4 Post-Configuration Verification & Conclusion

Following routing protocol convergence, routing tables populated correctly with remote subnets. ICMP reachability tests were repeated across end devices.

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC0	PC1	ICMP	
	Successful	PC2	PC1	ICMP	
	Successful	PC3	PC1	ICMP	
	Successful	PC0	PC3	ICMP	
	Successful	PC2	PC3	ICMP	

Figure 4: Successful ICMP ping verification showing full end-to-end network reachability post-RIPv2 deployment.

4.1 Conclusion

The successful convergence of RIPv2 verified that:

1. Disabling auto-summary (`no auto-summary`) prevents route ambiguity across sub-netted Class A/B boundaries.
2. Utilizing passive interface defaults restricts routing update broadcasts exclusively to inter-router links, reducing unnecessary network overhead.
3. Full end-to-end ICMP communication was restored across all subnets in the topology.